

CHEM 101 Topic 4: Atomic Structure — Answer Key

Section 1: History of Atomic Models

Problem 1: Thomson: Positive sphere with electrons; Rutherford: Dense positive nucleus. Discovery: Gold foil experiment.

Problem 2: Bohr used fixed orbits; Quantum model uses probability clouds (orbitals).

Problem 3: John Dalton; based on precise mass measurements of reactions.

Section 2: Atomic Notation and Mass

Problem 4: Chlorine (Cl); Protons: 17, Electrons: 17, Neutrons: 18.

Problem 5: Same protons/electrons, different neutrons (35 vs 37).

Problem 6: Mass Number = 56.

Section 3: Energy Levels & Orbitals

Problem 7: Total: 8 (2s: 2, 2p: 6).

Problem 8: Energy level: 3; Sublevel: d.

Problem 9: Aufbau Principle (lowest energy levels fill first).

Section 4: Electron Configurations

Problem 10: $1s^2 2s^2 2p^6 3s^2$

Problem 11: $1s^2 2s^2 2p^6 3s^2 3p^4$

Problem 12: $1s^2 2s^2 2p^6 3s^2 3p^6$